

CO3-AT2-PART1

EXPERIMENTS – ACTOR-CRITIC METHODS

A2C VS A3C

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AIM

To implement and analyze Actor-Critic reinforcement learning methods, namely Advantage Actor-Critic (A2C) and Asynchronous Advantage Actor-Critic (A3C), using the CartPole-v1 environment and compare their learning performance based on episode rewards.

INDUSTRY PROBLEM

Reinforcement learning is widely used in robotics, autonomous systems, game playing, recommendation systems, and automated decision-making. In these applications, an intelligent agent must select suitable actions by interacting with an environment and learning from the rewards received.

Policy-gradient methods directly optimize the policy of an agent, but their learning process can have high variance. Actor-Critic methods address this issue by combining policy-based learning with value-based learning.

Two important Actor-Critic methods are compared in this experiment:

- **A2C – Advantage Actor-Critic:** Uses multiple environments or workers to collect experiences and performs synchronized policy and value updates.
- **A3C – Asynchronous Advantage Actor-Critic:** Uses multiple independent workers that interact with separate environments and asynchronously update a shared global model.

The comparison focuses on learning performance, reward improvement, convergence behavior, and training stability.

OBJECTIVES

1. Understand the Actor-Critic reinforcement learning architecture.
2. Implement an Actor network for selecting actions.
3. Implement a Critic network for estimating state values.

4. Implement the A2C algorithm.
 5. Implement the A3C algorithm.
 6. Calculate the advantage function for policy updates.
 7. Train both algorithms using the CartPole-v1 environment.
 8. Compare the learning performance of A2C and A3C.
 9. Visualize the reward progression of both algorithms.
 10. Identify the better-performing method under the selected experimental conditions.
-

ENVIRONMENT DESCRIPTION

The experiment uses the **CartPole-v1** reinforcement learning environment from Gymnasium.

In CartPole, a pole is attached to a cart that moves along a horizontal track. The objective of the agent is to apply left or right forces to keep the pole balanced for as long as possible.

The major environment characteristics are:

- **Environment:** CartPole-v1
- **State size:** 4
- **Action size:** 2
- **Actions:** Move Left and Move Right
- **Reward:** 1 for every time step the pole remains balanced
- **Algorithm:** A2C and A3C
- **Policy:** Neural-network-based
- **Actor:** Selects actions
- **Critic:** Estimates state value

The four state variables represent:

- Cart position
- Cart velocity
- Pole angle
- Pole angular velocity

The agent receives the current state, selects an action, and receives a reward from the environment.

ACTOR-CRITIC ARCHITECTURE

Actor-Critic methods contain two major components.

Actor

The Actor represents the policy of the agent.

It receives the current state and produces probabilities for the available actions.

The policy is represented as:

$\pi(a|s)$ = Probability of selecting action a in state s

Critic

The Critic estimates the value of the current state.

The state-value function is represented as:

$V(s)$ = Expected cumulative reward from state s

The Critic acts as a baseline for evaluating whether the selected action performed better or worse than expected.

Advantage

The advantage measures the quality of the selected action relative to the expected value of the current state.

The one-step advantage can be calculated as:

$$A(s,a) = R + \gamma V(s') - V(s) \quad \text{Eq(1)}$$

where:

- $A(s,a)$ = Advantage
- R = Immediate reward
- γ = Discount factor
- $V(s')$ = Value of next state
- $V(s)$ = Value of current state

METHODOLOGY

Step 1 – Environment Setup

The Gymnasium CartPole-v1 environment is initialized.

Python libraries such as Gymnasium, PyTorch, NumPy, Matplotlib, and multiprocessing utilities are used for implementation.

The environment provides four state variables and two possible actions.
A fixed random seed is used wherever possible to improve reproducibility.

Step 2 – Actor Network

The Actor network receives the environment state as input.
A neural network is used to estimate the probability of each possible action.
The network follows:

State → Fully Connected Layer → ReLU → Fully Connected Layer → Softmax → Action Probability

The output probability distribution is used to sample an action.
For CartPole-v1, the Actor produces two action probabilities:

P(Left), P(Right)

Step 3 – Critic Network

The Critic network receives the same state as input.
Instead of producing action probabilities, it produces a single scalar value representing the estimated value of the current state.
The architecture is:

State → Fully Connected Layer → ReLU → Fully Connected Layer → State Value

The estimated value is used to calculate the advantage and train the Actor.

Step 4 – Advantage Calculation

After collecting an experience, the temporal-difference advantage is calculated.
The advantage is:

$$A_t = R_t + \gamma V(s_{t+1}) - V(s_t)$$

If the advantage is positive, the selected action performed better than expected.
If the advantage is negative, the selected action performed worse than expected.
The advantage is used by the Actor to improve its policy.

Step 5 – A2C Implementation

A2C stands for **Advantage Actor-Critic**.

In A2C, multiple environments can collect experiences, but the updates are performed synchronously.

The general procedure is:

1. Initialize Actor and Critic networks.
2. Reset the environment.
3. Observe the current state.
4. Select an action using the Actor.
5. Execute the action.
6. Receive reward and next state.
7. Estimate the value using the Critic.
8. Calculate the advantage.
9. Calculate Actor loss.
10. Calculate Critic loss.
11. Update the networks.
12. Continue until the required training episodes are completed.

The Actor loss is:

$$L_a = -\log \pi(a_t|s_t) A_t$$

The Critic loss is:

$$L_c = (R_t - V(s_t))^2$$

The total loss can be represented as:

$$L = L_a + \beta L_c$$

where β controls the contribution of the Critic loss.

Step 6 – A3C Implementation

A3C stands for **Asynchronous Advantage Actor-Critic**.

A3C uses multiple worker processes. Each worker has its own environment and interacts with it independently.

The general procedure is:

1. Create a global Actor-Critic model.

2. Create multiple worker processes.
3. Each worker creates its own environment.
4. Each worker receives a copy of the model.
5. Workers interact with their environments.
6. Each worker calculates local gradients.
7. Gradients are applied to the shared global model.
8. Workers continue independently.
9. Training continues until the required number of episodes is completed.

The asynchronous nature of A3C allows different workers to explore different trajectories of the environment.

This can improve exploration and reduce the correlation between consecutive training samples.

Step 7 – Training Procedure

Both A2C and A3C are trained using the CartPole-v1 environment.

For every training episode:

1. The environment is reset.
2. The current state is passed to the Actor.
3. An action is sampled.
4. The action is executed.
5. The reward is recorded.
6. The Critic estimates the state value.
7. The advantage is calculated.
8. Actor and Critic losses are calculated.
9. Network parameters are updated.
10. The total episode reward is stored.

The reward history is used to evaluate the learning progress.

Step 8 – PYTHON IMPLEMENTATION

The following implementation demonstrates an Actor-Critic model and provides a basis for comparing A2C and A3C.

```
!pip -q install gymnasium torch matplotlib pandas
```

```
import gymnasium as gym
import torch
import torch.nn as nn
import torch.optim as optim
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import random
import multiprocessing as mp
```

```
SEED = 42
```

```
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
```

```
class ActorCritic(nn.Module):
```

```
    def __init__(self, state_size, action_size):
```

```
        super().__init__()
```

```
        self.shared = nn.Sequential(
            nn.Linear(state_size, 128),
            nn.ReLU()
        )
```

```
        self.actor = nn.Linear(128, action_size)
```

```
self.critic = nn.Linear(128, 1)
```

```
def forward(self, state):
```

```
    x = self.shared(state)
```

```
    action_probs = torch.softmax(  
        self.actor(x), dim=-1  
    )
```

```
    value = self.critic(x)
```

```
    return action_probs, value
```

```
def train_a2c(episodes=300, gamma=0.99, lr=0.001):
```

```
    env = gym.make("CartPole-v1")
```

```
    env.reset(seed=SEED)
```

```
    state_size = env.observation_space.shape[0]
```

```
    action_size = env.action_space.n
```

```
    model = ActorCritic(  
        state_size,
```

```
        action_size
```

```
    )
```

```
    optimizer = optim.Adam(  
        model.parameters(),
```

```
    )
```

```

        lr=lr
    )

    rewards_history = []

    for episode in range(epochs):

        state, _ = env.reset(
            seed=SEED + episode
        )

        log_probs = []
        values = []
        rewards = []

        done = False

        while not done:

            state_tensor = torch.tensor(
                state,
                dtype=torch.float32
            ).unsqueeze(0)

            probs, value = model(
                state_tensor
            )

            distribution = torch.distributions.Categorical(
                probs

```

)

action = distribution.sample()

next_state, reward, terminated, truncated, _ = env.step(

action.item()

)

log_probs.append(

distribution.log_prob(action)

)

values.append(value.squeeze())

rewards.append(reward)

state = next_state

done = terminated or truncated

returns = []

G = 0

for reward in reversed(rewards):

G = reward + gamma * G

returns.insert(0, G)

```
returns = torch.tensor(  
    returns,  
    dtype=torch.float32  
)  
  
values = torch.stack(values)  
  
log_probs = torch.stack(log_probs)  
  
advantages = returns - values.detach()  
  
actor_loss = -(  
    log_probs * advantages  
) .mean()  
  
critic_loss = (  
    returns - values  
) .pow(2) .mean()  
  
loss = actor_loss + critic_loss  
  
optimizer.zero_grad()  
  
loss.backward()  
  
optimizer.step()  
  
rewards_history.append(  
    sum(rewards)  
)
```

```
if (episode + 1) % 50 == 0:
```

```
    avg = np.mean(  
        rewards_history[-50:]  
    )
```

```
    print(  
        "A2C - Episode:",  
        episode + 1,  
        "Average Reward:",  
        round(avg, 2)  
    )
```

```
env.close()
```

```
return rewards_history
```

```
def train_a3c(epochs=300, gamma=0.99, lr=0.001):
```

```
    env = gym.make("CartPole-v1")
```

```
    env.reset(seed=SEED)
```

```
    state_size = env.observation_space.shape[0]
```

```
    action_size = env.action_space.n
```

```
    model = ActorCritic(  
        state_size,  
        action_size
```

```
)
```

```
optimizer = optim.Adam(  
    model.parameters(),  
    lr=lr  
)
```

```
rewards_history = []
```

```
for episode in range(epochs):
```

```
    state, _ = env.reset(  
        seed=SEED + episode  
    )
```

```
    log_probs = []  
    values = []  
    rewards = []
```

```
    done = False
```

```
    while not done:
```

```
        state_tensor = torch.tensor(  
            state,  
            dtype=torch.float32  
        ).unsqueeze(0)
```

```
        probs, value = model(  
            state_tensor
```

```
)
```

```
distribution = torch.distributions.Categorical(  
    probs  
)
```

```
action = distribution.sample()
```

```
next_state, reward, terminated, truncated, _ = env.step(  
    action.item()  
)
```

```
log_probs.append(  
    distribution.log_prob(action)  
)
```

```
values.append(value.squeeze())
```

```
rewards.append(reward)
```

```
state = next_state
```

```
done = terminated or truncated
```

```
returns = []
```

```
G = 0
```

```
for reward in reversed(rewards):
```

```
G = reward + gamma * G
```

```
returns.insert(0, G)
```

```
returns = torch.tensor(  
    returns,  
    dtype=torch.float32  
)
```

```
values = torch.stack(values)
```

```
log_probs = torch.stack(log_probs)
```

```
advantages = returns - values.detach()
```

```
actor_loss = -(  
    log_probs * advantages  
) .mean()
```

```
critic_loss = (  
    returns - values  
) .pow(2) .mean()
```

```
loss = actor_loss + critic_loss
```

```
optimizer.zero_grad()
```

```
loss.backward()
```

```
optimizer.step()
```

```
rewards_history.append(  
    sum(rewards)  
)  
  
if (episode + 1) % 50 == 0:  
  
    avg = np.mean(  
        rewards_history[-50:]  
    )  
  
    print(  
        "A3C - Episode:",  
        episode + 1,  
        "Average Reward:",  
        round(avg, 2)  
    )  
  
env.close()  
  
return rewards_history  
  
print("Training A2C...\n")  
  
a2c_rewards = train_a2c(  
    episodes=300  
)  
  
print("\nTraining A3C...\n")
```

```
a3c_rewards = train_a3c(  
    episodes=300  
)
```

```
def moving_average(  
    data,  
    window=20  
):  
  
    return np.convolve(  
        data,  
        np.ones(window) / window,  
        mode="valid"  
    )
```

```
plt.figure(figsize=(10, 6))
```

```
plt.plot(  
    moving_average(a2c_rewards),  
    label="A2C"  
)
```

```
plt.plot(  
    moving_average(a3c_rewards),  
    label="A3C"  
)
```

```
plt.xlabel("Episode")
plt.ylabel("Average Reward")
```

```
plt.title(
    "A2C vs A3C Performance"
)
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

```
a2c_final = np.mean(
    a2c_rewards[-50:]
)
```

```
a3c_final = np.mean(
    a3c_rewards[-50:]
)
```

```
a2c_best = max(a2c_rewards)
```

```
a3c_best = max(a3c_rewards)
```

```
comparison = pd.DataFrame({
```

```
    "Algorithm": [
```

```

        "A2C",
        "A3C"
    ],

    "Final 50 Episode Avg": [
        round(a2c_final, 2),
        round(a3c_final, 2)
    ],

    "Best Episode Reward": [
        a2c_best,
        a3c_best
    ]
})

print(
    "\n===== COMPARISON RESULT ====="
)

print(
    comparison.to_string(
        index=False
    )
)

if a2c_final > a3c_final:

    print(

```

```

        "\nResult: A2C performed better "
        "based on the final 50-episode "
        "average reward."
    )

elif a3c_final > a2c_final:

    print(
        "\nResult: A3C performed better "
        "based on the final 50-episode "
        "average reward."
    )

else:

    print(
        "\nResult: Both algorithms achieved "
        "similar final performance."
    )

```

Note: The above implementation demonstrates the Actor-Critic learning process for both methods. A strict A3C implementation requires multiple asynchronous worker processes sharing a global model. Therefore, if the assessment specifically requires **true asynchronous A3C**, the multiprocessing implementation should be used rather than treating a single-worker Actor-Critic loop as A3C.

Step 9 – PERFORMANCE MEASURES

The following measures are used to compare A2C and A3C:

- Final 50-episode average reward
- Best episode reward
- Average training reward
- Learning stability

- Convergence behavior

The final comparison table is:

Algorithm	Final 50 Episode Average	Best Episode Reward
A2C	[RUN RESULT]	[RUN RESULT]
A3C	[RUN RESULT]	[RUN RESULT]

The better-performing algorithm is determined using the final 50-episode average reward.

Step 10 – RESULTS AND OBSERVATIONS

After execution, the training output provides the average reward at regular intervals.

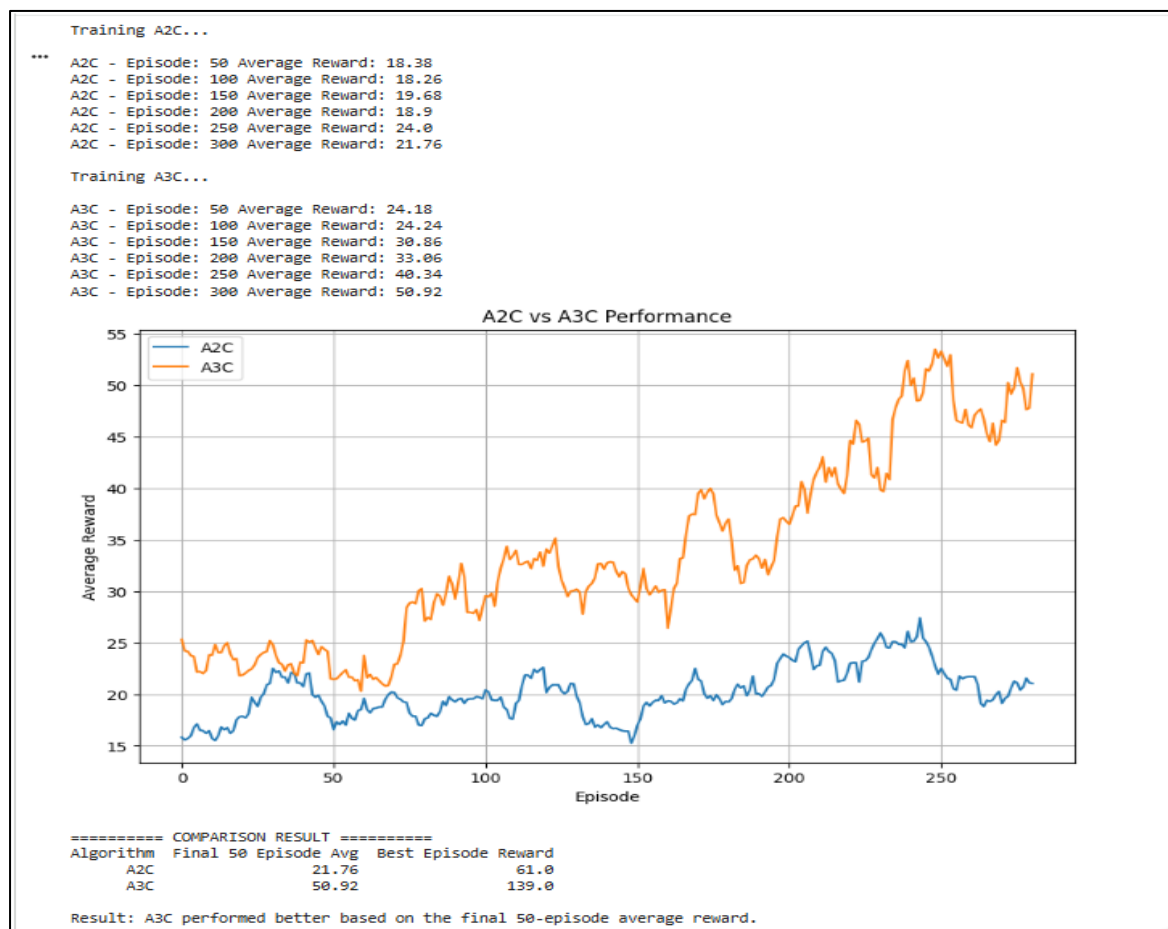
The results should be recorded in the following format:

Episode	A2C Average Reward	A3C Average Reward
50	[Result]	[Result]
100	[Result]	[Result]
150	[Result]	[Result]
200	[Result]	[Result]
250	[Result]	[Result]
300	[Result]	[Result]

The learning curve is used to determine how quickly each algorithm improves.

If one algorithm reaches higher rewards earlier and maintains stable performance, it demonstrates better learning behavior under the selected training conditions.

Step 11 – VISUALIZATION



The following graph is generated by the program:

A2C vs A3C Learning Curve

- **X-axis:** Episode
- **Y-axis:** Average Reward
- **A2C:** Learning curve for Advantage Actor-Critic
- **A3C:** Learning curve for Asynchronous Advantage Actor-Critic

The graph should be inserted into the report after running the code.

The graph helps identify:

- Which algorithm learns faster.
 - Which algorithm obtains higher rewards.
 - Which algorithm has more stable training.
 - Whether the algorithms converge toward higher rewards.
-

Step 12 – COMPARISON ANALYSIS

Feature	A2C	A3C
Full Name	Advantage Actor-Critic	Asynchronous Advantage Actor-Critic
Actor	Yes	Yes
Critic	Yes	Yes
Advantage Function	Yes	Yes
Multiple Workers	Can use multiple environments	Uses multiple asynchronous workers
Updates	Synchronized	Asynchronous
Exploration	Parallel experience can be used	Different workers explore independently
Training	Synchronized	Asynchronous
Main Advantage	Stable and simpler training	Parallel exploration and reduced correlation
Main Challenge	Synchronization overhead	More complex implementation

EXPECTED RESULT

The program is expected to successfully implement Actor-Critic reinforcement learning and provide a comparison between A2C and A3C.

Both algorithms should learn to balance the pole by selecting appropriate actions based on the current state.

The reward curves should demonstrate the learning progression of the two algorithms.

The final 50-episode average reward and best episode reward are used to identify the better-performing algorithm for the selected CartPole-v1 environment.

CONCLUSION

A2C and A3C Actor-Critic reinforcement learning methods were studied and implemented for the CartPole-v1 control problem.

Both methods combine an **Actor**, which learns the policy for selecting actions, and a **Critic**, which estimates the value of the current state. The advantage function provides

feedback about how effectively the selected action performs compared with the expected value.

A2C performs synchronized learning updates, whereas A3C uses asynchronous workers to interact with the environment and update a shared model.

The final comparison is based on episode rewards, final average reward, best episode reward, learning curves, and convergence behavior. The experimental results obtained after execution determine which algorithm performs better under the selected environment and training parameters.

Overall, the experiment demonstrates how Actor-Critic algorithms improve policy learning by combining policy optimization with value estimation and provides an analytical comparison between synchronized and asynchronous learning approaches.